

Amal Shiyas

Mechanical Engineering Graduate

✉ amalshiyasabdulrahman@gmail.com ☎ 9037101027 📍 Kollam 🌐 Amal Shiyas 📁 Portfolio

PROFILE

Mechanical Engineering graduate with practical exposure to hydraulic assembly, manufacturing, inspection, and testing. Experienced in mechanical CAD modeling, engineering drawings, and prototype development through academic and personal projects. Uses Autodesk Fusion 360 and has experience with SolidWorks, AutoCAD, 3D printing, and Arduino. Interested in mechanical design, automotive engineering, and manufacturing.

PROJECTS

Real time Spatio-Temporal pressure mapping in contact mechanics 2026

Final-year team project | Fusion 360, 3D Printing, Arduino, Python

- Designed and assembled a prototype integrating a 3D-printed enclosure, Velostat sensing matrix, Arduino, and supporting electronics.
- Conducted single-point and distributed-load tests to evaluate pressure response and real-time data acquisition.
- Contributed to redesigning the initial FSR prototype to address ghost readings, sensor interference, and wiring complexity; collaborated on testing and Python heat-map visualization.

Toyota Corolla Front Brake Rotor 2026

Autodesk Fusion 360

- Designed a reference-based parametric CAD model incorporating hub geometry, center bore, 5-lug bolt pattern, chamfers, and fillets using Fusion 360.
- Developed a dimensioned **2D engineering drawing** and photorealistic CAD renders for technical documentation and visualization.

Hand gesture control Wheelchair prototype 2025

Built a gesture-controlled wheelchair prototype that translates hand movements into real-time directional control, enabling intuitive mobility without traditional input devices.

EXPERIENCE

Wipro Infrastructure Engineering 06/2026 – 07/2026 Apprentice Engineer Chennai, India

- Working in the Hydraulics division supporting assembly and manufacturing operations.
- Assisting in the assembly, inspection, and testing of hydraulic components and systems.
- Following industrial quality standards and safety procedures during production activities.

United Electrical Industries Limited. 04/2025 – 05/2025 Intern Kollam, Kerala

- Gained hands-on exposure to industrial manufacturing processes, including machining, assembly, welding and electroplating, improving my understanding of real factory operations and product development.
- Learned quality control and testing methods for electrical components, strengthening my skills in inspection, evaluation, and standards compliance.

Kerala Minerals Metals Ltd 🌐 08/2024 – 09/2024 Intern Kollam, India

- Gained hands-on experience in titanium dioxide manufacturing, from mineral separation to final product
- Studied the chloride route process for TiO₂ production and synthetic rutile manufacturing
- Observed industrial-scale mineral processing operations and quality control procedures

Kerala Electrical & Allied Engineering Co.Ltd. 🌐 10/2023 – 11/2023 Intern Kollam, India

- Applied mechanical engineering concepts in practical industrial settings at KEL (Kerala Electrical & Allied Engineering Co.Ltd 🌐)
- Collaborated with industry professionals on day-to-day operations and maintenance procedures
- Gained practical exposure to industrial electrical and mechanical systems

SKILLS

Mechanical Design & Prototyping: SolidWorks, AutoCAD, Fusion 360, 3D Printing, Engineering Drawing Interpretation, basic knowledge of GD&T (geometric dimensioning and tolerancing)

Testing & Validation: Experimental Testing, Sensor Integration, Arduino, Data Acquisition

Analysis: ANSYS, MATLAB

Manufacturing: Mechanical Assembly, Manufacturing Processes, Quality Control

EDUCATION

Mechanical Engineering

Amrita Vishwa Vidyapeetham [@](#)

CGPA- 6.82/10

10/2022 – 2026

Kollam, India

High School and Higher Secondary

Stratford Public School [@](#)

12th with 78.9%

10th with 89.8%

2022

Kollam, India

CERTIFICATIONS

Computational Fluid Mechanics - Airflow Around a Spoiler [@](#)

07/2025

Coursera

- Understand cloud-simulation technology and why it is so powerful.
- How to set up an incompressible fluid flow simulation around a spoiler and post-process the results.

AUTOMOTIVE & STUDENT ACTIVITIES

IKR-Indian Kart Racing 2025

Participated in IKR 2025 (Indian Kart Racing) held at Coastt Speedway under ISIE – Imperial Society of Innovative Engineers, as part of Team Asphalt Racing. Contributed to design, fabrication, and racing activities, with the team securing awards in the Lightest Weight Category and achieving an All-India Rank of 13 among participating teams.

Executive Member

ASME AVV

Executive Member, ASME (American Society of Mechanical Engineers)

Actively contributed to organizing technical events, workshops, and competitions aimed at enhancing technical knowledge and practical skills among students.

Collaborated with peers to foster a culture of innovation and engineering excellence.

Kollam, India

VOLUNTEERING

Experience as a Scribe

Served as a scribe for a student in my second year of college, assisting them during their university exams. This role enhanced my interpersonal communication skills and taught me how to effectively and sensitively communicate with others, while further nurturing my commitment to service.

Community Service

A Student Social Responsibilities initiative

Workshop Volunteer – Health and Wellbeing Lourde Mata School, Chavara

- Conducted an engaging workshop for school students on health and wellbeing.
- Helped raise awareness about healthy habits and lifestyle choices among young learners.
- Developed communication and presentation skills through interactive sessions.

Amritavarsham 2025

Volunteered at Amritavarsham celebrations in Amritapuri, supporting event coordination, crowd management, and logistics, ensuring smooth operations during large-scale cultural and spiritual programs. Gained hands-on experience in teamwork, communication, and handling high-responsibility tasks in a fast-paced environment.

LANGUAGES

English

Malayalam